Discrete - 1.1.63

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Question

Quadrature formula: $\int_0^1 f(x)\,dx \approx \frac{1}{12}[f(0) + 2bf(0.25) + 2f(0.5) + 2df(0.75) + f(1)]$.

If used as Simpson's $\frac{1}{3}$ rule: (2007 XE)

a) b = d = 1 b) b = d = 2 c) b = 2d = 1 d) b = 2d = 2

Solution :

Simpson's 1/3 Rule

Simpson's $1/3$ rule is a numerical integration method that approximates the area under a curve by fitting parabolic arcs (second-order polynomials) through sets of three points.

For a single interval $[x_0, x_2]$ with a midpoint x_1 and step size h:

$$\int_{x_0}^{x_2} f(x)\,dx \approx \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right] \tag{1}$$

In general Simpson's 1/3 Rule over n intervals (where n must be even):

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[f(x_0) + 4\sum_{i=1,3,5}^{n-1} f(x_i) + 2\sum_{j=2,4,6}^{n-2} f(x_j) + f(x_n)\right] \tag{2}$$

Where $h = \frac{b-a}{n}$ is the uniform step size and x_0 = a, x_1 = b.

The problem provides a specific quadrature formula:

$$\int_0^1 f(x)\,dx \approx \frac{1}{12}\left[f(0) + 2bf(0.25) + 2f(0.5) + 2df(0.75) + f(1)\right] \tag{3}$$

Based on the function arguments $\{0, 0.25, 0.5, 0.75, 1\}$:

- $a = 0, b = 1$
- $n = 4$ intervals
- $h = \frac{1-0}{4} = 0.25$

Substituting $h = 0.25$ into the general Simpson's Rule for $n = 4$:

$$\int_0^1 f(x)\,dx \approx \frac{0.25}{3}\left[f(0) + 4f(0.25) + 2f(0.5) + 4f(0.75) + f(1)\right] \tag{4}$$

Simplifying the scalar $\frac{0.25}{3} = \frac{1}{12}$, the standard internal weights are:

$$\text{Internal Weights} = [4, 2, 4] \tag{5}$$

Comparing these to the coefficients inside the brackets of the equation (3) we get

- Coefficient for $f(0.25)$: $2b = 4 \implies \mathbf{b = 2}$
- Coefficient for $f(0.75)$: $2d = 4 \implies \mathbf{d = 2}$

Numerical Example
Lets test the above relation with $f(x) = x$
Exact Integral :

$$\int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2} = 0.5 \tag{6}$$

Applying the Formula

$$\int_0^1 f(x)\,dx \approx \frac{0.25}{3}\left[f(0) + 4f(0.25) + 2f(0.5) + 4f(0.75) + f(1)\right] \tag{7}$$

- $f(0) = 0$
- $f(0.25) = 0.25 = \frac{1}{4}$
- $f(0.5) = 0.5 = \frac{1}{2}$
- $f(0.75) = 0.75 = \frac{3}{4}$
- $f(1) = 1$

Substitute these into above equation and simplifying it will give :

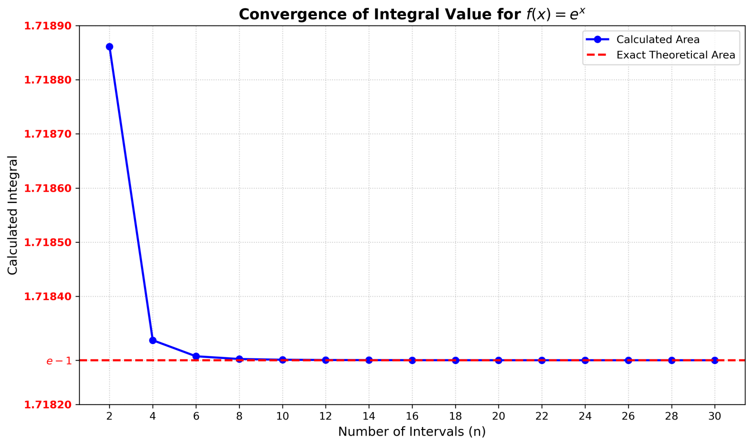
$$\frac{1}{12}\left[0 + 4\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) + 4\left(\frac{3}{4}\right) + 1\right] \tag{8}$$

$$\implies \frac{1}{12}\left[1 + 1 + 3 + 1\right] \tag{9}$$

$$\implies \frac{6}{12} = \frac{1}{2} \tag{10}$$

- In the case of linear and quadratic functions simpson′s $\frac{1}{3}$ rule gives the perfect answer at the first attempt only i.e n = 2.
- This is because simpson′s $\frac{1}{3}$ rule fits parabolas.
- Let′s check for different functions like $f(x) = e^x$

$f(x) = e^x$:



$f(x) = e^x$ integral convergence

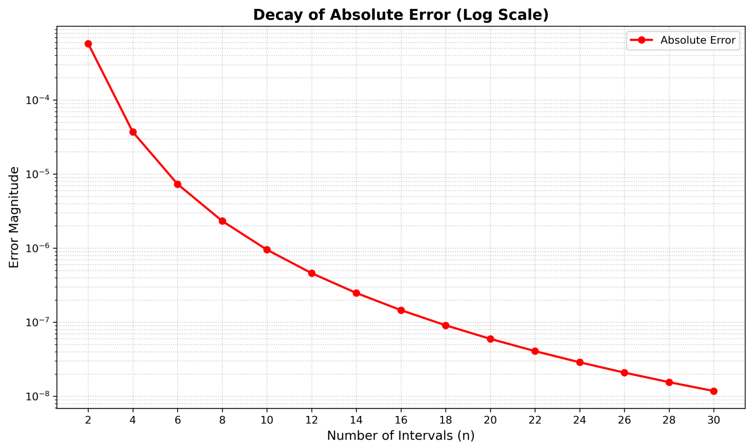
$$\int_0^1 e^x \, dx = [e^x]_0^1 \tag{11}$$

$$= e^1 - e^0 \tag{12}$$

$$= e - 1 = 1.71828 \tag{13}$$

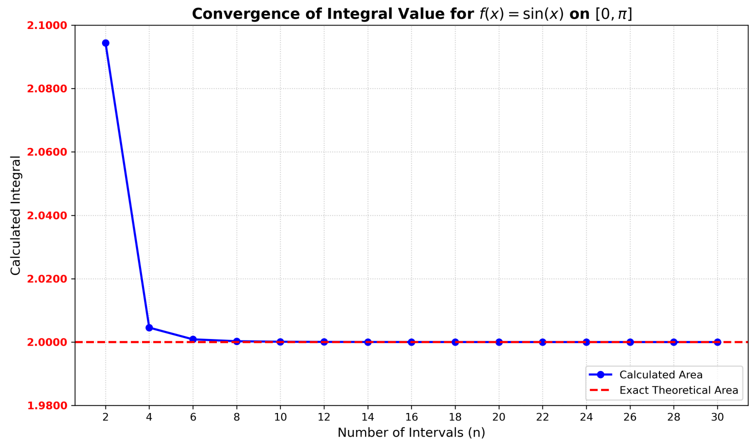
- From the above graph it is very clear that for the function $f(x) = e^x$ the value of integral computed using simpson's $\frac{1}{3}$ rule coincides with the actual integral from n = 6

Error :



$f(x) = e^x$ error decay

$f(x) = sin(x)$:



$f(x) = e^x$ integral convergence

$$\int_0^{\pi} \sin(x)\,dx = [-\cos(x)]_0^{\pi} \tag{14}$$

$$= (-\cos(\pi)) - (-\cos(0)) \tag{15}$$

$$= (-(-1)) - (-1) \tag{16}$$

$$= 1 + 1 \tag{17}$$

$$= 2 \tag{18}$$

- From the above graph it is very clear that for the function $f(x) = sin(x)$ the value of integral computed using simpson's $\frac{1}{3}$ rule coincides with the actual integral from n = 6

Error :

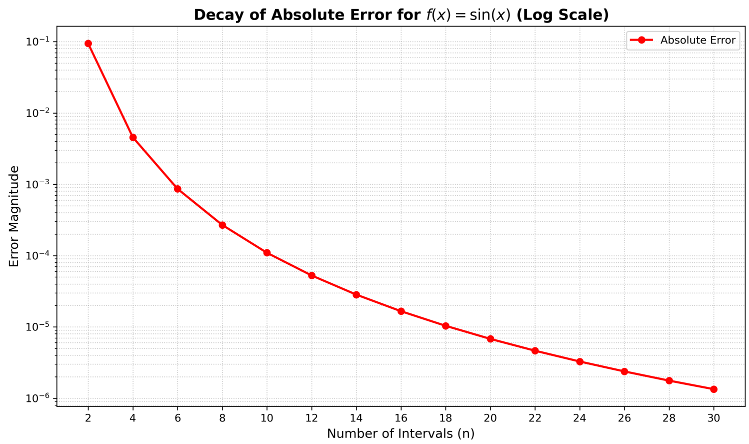


Figure 0: $f(x) = sin(x)$ error decay